

# Référence CIS Red Hat Enterprise Linux 10

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# Overview

All CIS Benchmarks™ (Benchmarks) focus on technical configuration settings used to maintain and/or increase the security of the addressed technology, and they should be used in conjunction with other essential cyber hygiene tasks like:

- Monitoring the base operating system and applications for vulnerabilities and quickly updating with the latest security patches
- Endpoint protection (antivirus software, Endpoint Detection and Response (EDR), etc.)
- Logging and monitoring user and system activity

In the end, the Benchmarks are designed to be a key component of a comprehensive cybersecurity program.

## Important Usage Information

All Benchmarks are available free for non-commercial use from the [CIS Website](#). They can be used to manually assess and remediate systems and applications. In lieu of manual assessment and remediation, there are several tools available to assist with assessment:

- [CIS Configuration Assessment Tool \(CIS CAT Pro Assessment\)](#)
- [CIS Benchmarks™ Certified for Policy Testing](#)

These tools make the hardening process much more scalable for large numbers of systems and applications.

**NOTE:** Some testing focuses only on the Benchmark Recommendations that can be fully automated (labeled **Manual**). It is important that **All** Recommendations (**Automated** and **Manual**) be addressed since all are important for properly securing systems and are typically in scope for audits.

## Key Stakeholders

Cybersecurity is a collaborative effort, and cross functional cooperation is imperative within an organization to discuss, test, and deploy Benchmarks in an effective and efficient way. The Benchmarks are developed to be best practice configuration guidelines applicable to a wide range of use cases. In some organizations, exceptions to specific Recommendations will be needed, and the team should work to prioritize the problematic Recommendations based on several factors like risk, time, cost, and other. These exceptions should be properly categorized and documented for auditing purposes.

## Apply the Correct Version of a Benchmark

Benchmarks are developed and tested for a specific set of products and versions and applying an incorrect Benchmark to a system can cause the resulting assessed score to be incorrect. This is due to the assessment of settings that do not apply to the target system. To ensure the correct Benchmark is being assessed:

1. **Deploy the Benchmark applicable to the way settings are managed in the environment.** An example of this is the Microsoft Windows family of Benchmarks, which have separate Benchmarks for Group Policy, registry, and file share systems based upon how system management is deployed. Applying the wrong Benchmark in this case will give invalid results.
2. **Use the most recent version of a Benchmark.** This is true for all Benchmarks, but especially true for cloud technologies. Cloud technologies change frequently and using an older version of a Benchmark may have invalid methods for auditing and remediation.

## Exceptions

The guidance items in the Benchmarks are called recommendations and not requirements, and exceptions to some of them are expected and acceptable. The Benchmarks strive to be a secure baseline, or starting point, for a specific technology, with known issues identified during Benchmark development are documented in the Impact section of each Recommendation. In addition, organizational system specific requirements, or local site policy may require changes as well, or an exception to a Recommendation or group of Recommendations (e.g. A Benchmark could Recommend that a Web server not be installed on the system, but if a system's primary purpose is to function as a Webserver, there should be a documented exception to this Recommendation for that specific server).

In the end, exceptions to some Benchmark Recommendations are common and acceptable, and should be handled as follows:

1. The reasons for the exception should be reviewed cross functionally and be well documented for audit purposes.
2. A plan should be developed for mitigating, or eliminating, the exception in the future, if applicable.
3. If the organization decides to accept the risk of this exception and work toward mitigation or elimination, this should be documented for audit purposes.

It is the responsibility of the organization to determine their overall security policy, and which settings are applicable to their unique needs based on the overall risk profile for the organization.

## Remediation

CIS has developed [Build Kits](#) for many technologies to assist in the automation of hardening systems. Build Kits are designed to correspond to Benchmark's "Remediation" section, which provides the required remediation steps necessary to make the Recommendation compliant to the Benchmark.

**When remediating systems (changing configuration settings on deployed systems as per the Benchmark's Recommendations), please approach this with caution and test thoroughly.**

The following is a reasonable remediation approach to follow:

1. CIS Build Kits, or internally developed remediation methods should never be applied to production systems without proper testing.
2. Proper testing consists of the following:
  - a. Understand the configuration (including installed applications) of the targeted systems. Various parts of the organization may need different configurations (e.g., software developers vs. distant office workers).
  - b. Read the Impact section of the given Recommendation to help determine if there might be an issue with the targeted systems.
  - c. Test the configuration changes with representative test systems. If issues arise during testing, they can be resolved prior to deploying to any production systems.
  - d. When testing is complete, initially deploy to a small sub-set of production systems and monitor closely for issues. If there are issues, they can be resolved prior to deploying more broadly.
  - e. When the initial deployment shows to complete successfully, gradually deploy to additional systems and monitor closely for issues. Repeat this process until the full deployment is complete.

## Summary

Using the Benchmark's Certified tools, working as a team with key stakeholders, being proactive with exceptions, and being careful with remediation deployment, it is possible to handle large numbers of deployed systems in a cost-effective, efficient, and safe manner.

**NOTE:** As previously stated, the PDF versions of the CIS Benchmarks™ are available for free, non-commercial use on the [CIS Website](#). All other formats of the CIS Benchmarks™ (API, Word, Excel, and [Build Kits](#)) are available for CIS [Academy](#)™ members.

CIS iCIS™ Pro is also available to CIS [Academy](#)™ members.

## Target Technology Details

This document provides prescriptive guidance for establishing a secure configuration posture for Red Hat Enterprise Linux 10 systems running on x86\_64 platforms.

This guide was developed and tested against Red Hat Enterprise Linux 10.0.

The guidance within broadly assumes that operations are being performed as the `root` user, and executed under the default shell access for the applicable distribution. Operations performed using `sudo` instead of the `root` user, or executed under another shell, may produce unexpected results, or fail to make the intended changes to the system. Non-root users may not be able to access certain areas of the system, especially after remediation has been performed. It is advisable to verify `root` users' path integrity and the integrity of any programs being run prior to execution of commands and scripts included in this benchmark.

The default prompt for the `root` user is `#`, and as such all sample commands will have `#` as an additional indicator that it is to be executed as `root`.

To obtain the latest version of this guide, please visit <https://redhat.com/rhcs>. If you have questions, comments, or have identified ways to improve this guide, please write to us at [rhcs@redhat.com](mailto:rhcs@redhat.com).

## Intended Audience

This benchmark is intended for system and application administrators, security specialists, auditors, help desks, and platform deployment personnel who plan to develop, deploy, assess, or secure solutions that incorporate Red Hat Enterprise Linux 10 on x86\_64 platforms.